Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

Load Dataset

```
df = pd.read_csv('/content/Iris.csv')
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: Generate code with df New interactive sheet

Understand Dataset

```
print(df.columns)
print(df.info())
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
       'Species'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
None
```

Data Preprocessing

```
# Remove unnecessary column
df = df.drop(columns=['Id'])

# Use only numerical features (for clustering)
X = df[['SepalLengthCm', 'SepalWidthCm',
        'PetalLengthCm', 'PetalWidthCm']]
```

Feature Scaling

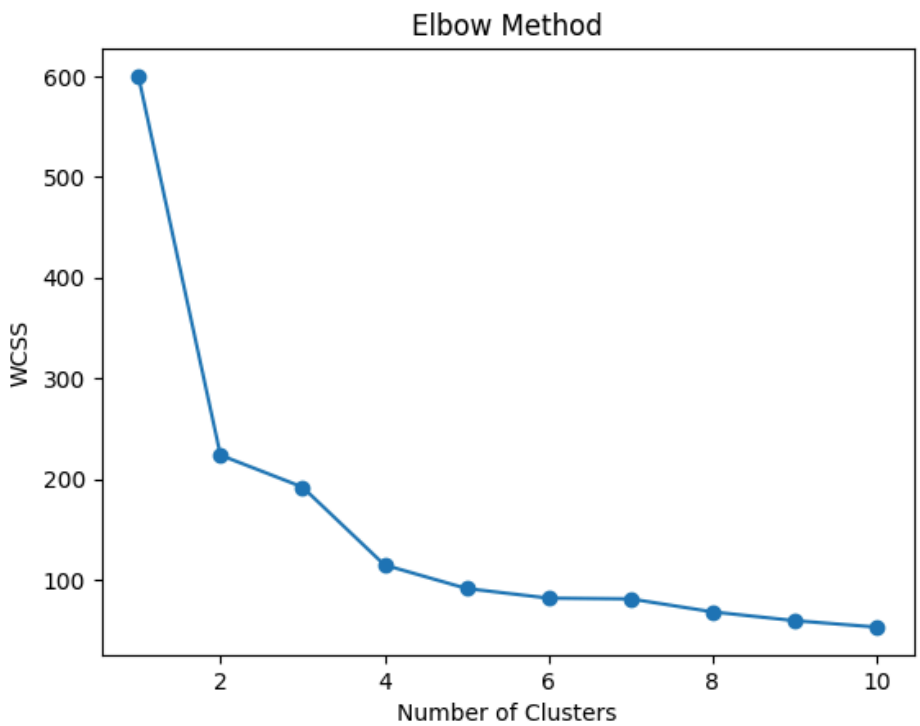
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

lbow Method

```
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```

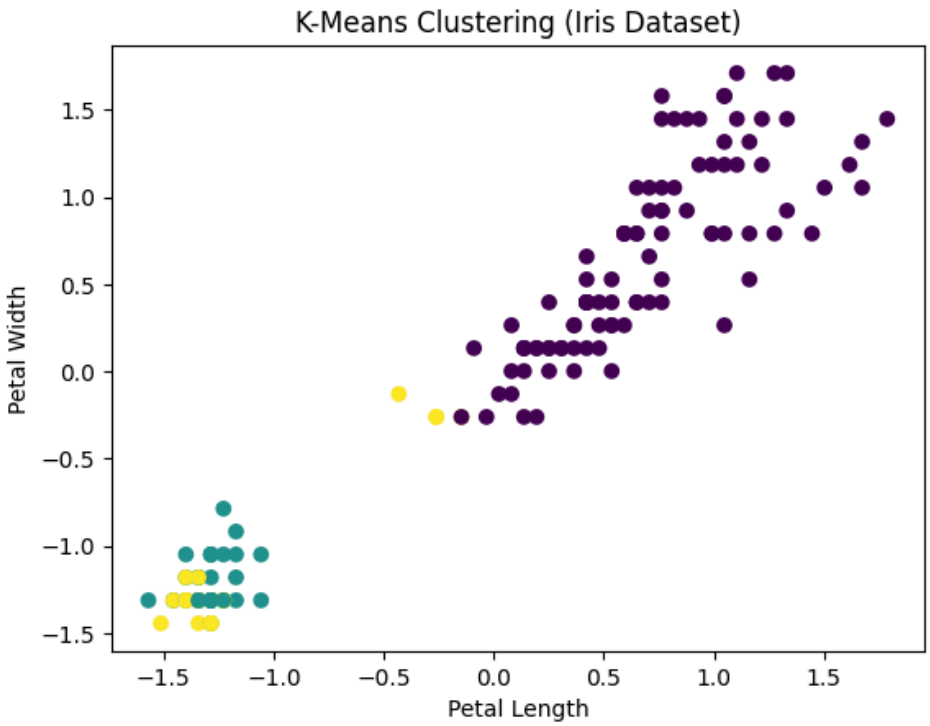


Apply K-Means

```
kmeans = KMeans(n_clusters=3, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)
```

Visualization

```
plt.scatter(X_scaled[:, 2], X_scaled[:, 3], c=y_kmeans)
plt.xlabel("Petal Length")
plt.ylabel("Petal Width")
plt.title("K-Means Clustering (Iris Dataset)")
plt.show()
```



Add Cluster Label

```
df['Cluster'] = y_kmeans
df.head()
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	Cluster
0	5.1	3.5	1.4	0.2	Iris-setosa	1
1	4.9	3.0	1.4	0.2	Iris-setosa	2
2	4.7	3.2	1.3	0.2	Iris-setosa	2
3	4.6	3.1	1.5	0.2	Iris-setosa	2
4	5.0	3.6	1.4	0.2	Iris-setosa	1

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Compare with Actual Species

```
print(pd.crosstab(df['Cluster'], df['Species']))
```

```
Species  Iris-setosa  Iris-versicolor  Iris-virginica
Cluster
0                  0               46              50
1                 32                0               0
2                 18                4               0
```